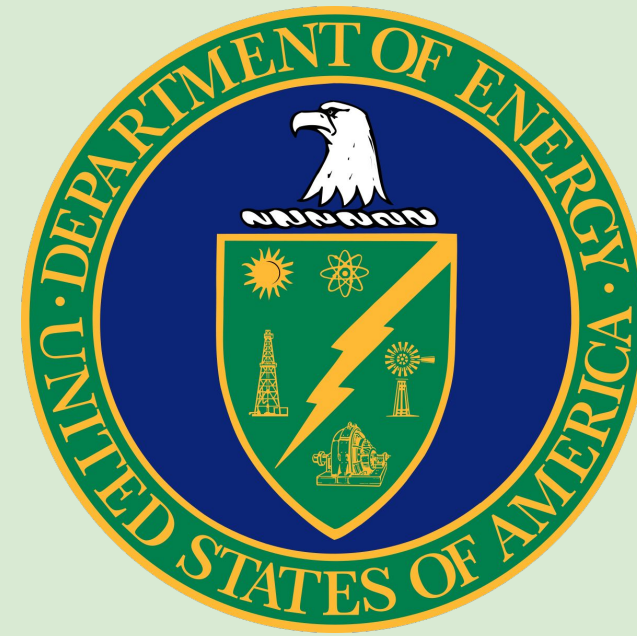




Geomancy: Automated Performance Enhancement through Data Layout Optimization

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Overview

- ❖ **Large distributed storage systems such as high performance computing (HPC) systems**
 - Must feature sufficient performance and scale for demanding scientific workloads
 - Must handle shifting workloads without causing bottlenecks
- ❖ **Data should be relocated preempting performance drops**
 - Large storage systems inhibits rapid reconstruction of data layout
 - Size and complexity
 - Large number of features used to track access performance
 - Prevent accurate modeling of performance
 - Increase modeling time overhead
- ❖ **Geomancy**
 - Automatically optimizes the placement of data within a distributed storage system
 - Leverages a neural network architecture
 - Accurately forecasts future performance of accesses
- ❖ **Benefits for storage systems such as**
 - Avoiding potential bottlenecks
 - Increasing overall I/O throughput by **11% to 30%**

Background and Motivation

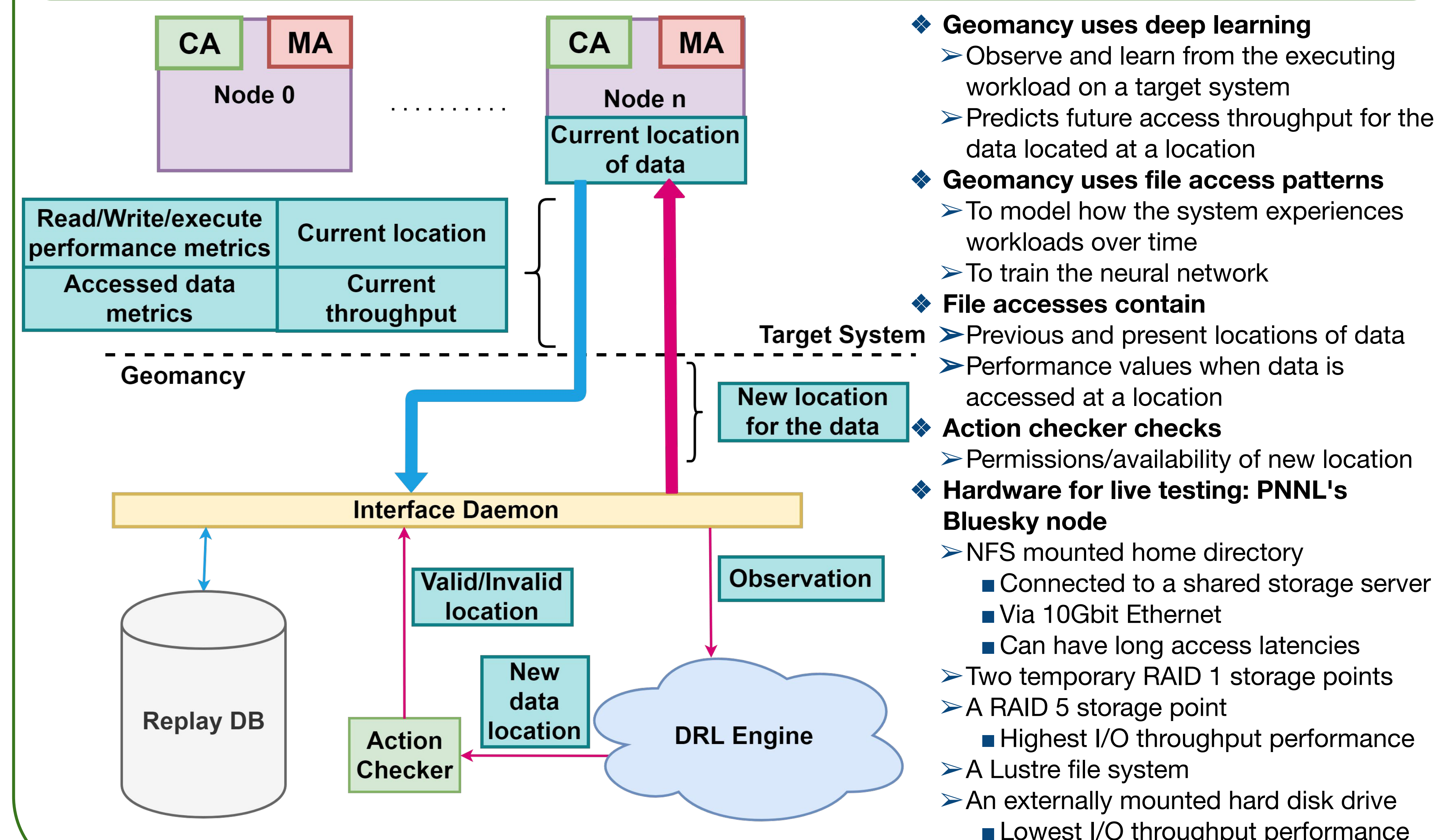
- ❖ **Moving the code to the data**
 - Achieved by spreading computational resources among many computation nodes
 - Hadoop
 - MapReduce
 - Twister
 - Problem: the workload is not storage bottlenecked
 - Physics workloads require hundreds of terabytes of data
 - Moving compute to the data requires specialized system architectures
- ❖ **Data layout strategies**
 - Based on the data distribution across storage devices
 - Heuristics cannot determine if a change in latency is long lasting
 - Limited by their inability to quickly search through enough data layout
- ❖ **Motivating workloads**
 - Traces generated from a Monte Carlo physics workload provided by Pacific Northwest National Lab (PNNL)
 - Utilizes data from the specialized particle detector
 - From the SuperKEKB collider in Japan
 - Workload traces from CERN
 - A storage cluster with an analysis, archive and tape pool
 - An analysis pool is a low latency disk base caching storage

Best Neural networks accuracy

Model Number	Mean of Absolute Error	Standard Deviation of Absolute Error
Model 1	18.88	16.92
Model 2	18.77	16.83
Model 3	17.63	15.95
Model 4	17.72	16.02
Model 5	18.50	16.42

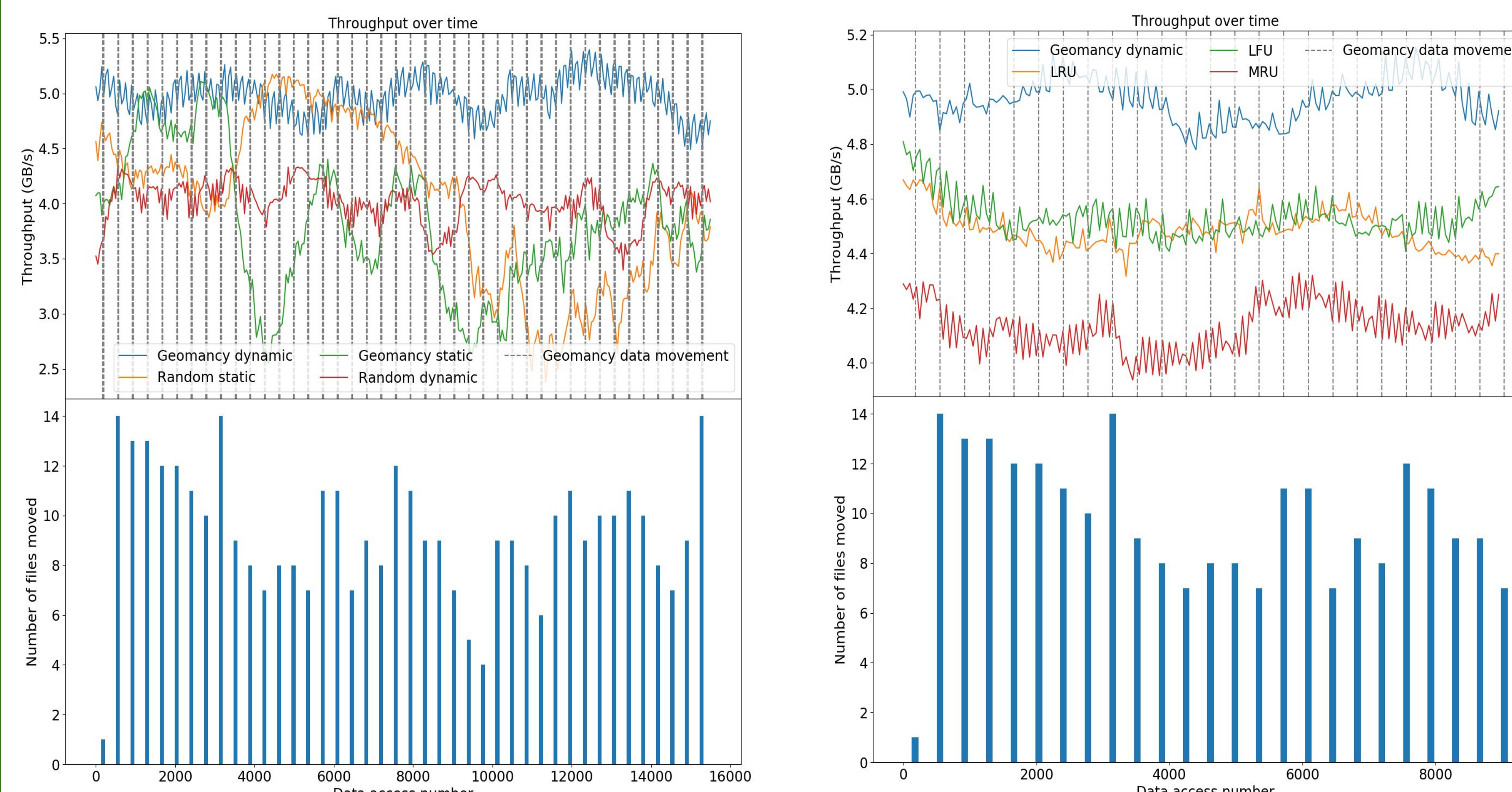
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The System



- ❖ **Geomancy uses deep learning**
 - Observe and learn from the executing workload on a target system
 - Predicts future access throughput for the data located at a location
- ❖ **Geomancy uses file access patterns**
 - To model how the system experiences workloads over time
 - To train the neural network
- ❖ **File accesses contain**
 - Previous and present locations of data
 - Performance values when data is accessed at a location
- ❖ **Action checker checks**
 - Permissions/availability of new location
- ❖ **Hardware for live testing: PNNL's Bluesky node**
 - NFS mounted home directory
 - Connected to a shared storage server
 - Via 10Gbit Ethernet
 - Can have long access latencies
 - Two temporary RAID 1 storage points
 - A RAID 5 storage point
 - Highest I/O throughput performance
 - A Lustre file system
 - An externally mounted hard disk drive
 - Lowest I/O throughput performance

Preliminary results



Future Work

- ❖ **Automatic feature selection**
 - Determine the combination of features that produces the most accurate predictions
 - Apply the updated predictions to the system
- ❖ **Preemptive data movement**
 - Preempting workload changes by changing locations of data depending on the time of day
 - Handling concurrent workloads accessing similar data

Extended Goals:

- ❖ **Concurrent access data movement**
 - Multiple instances of the PNNL workload running concurrently
 - Have a neural network to predict next file movement possibility
 - Large enough time gap with no access to a file.
- ❖ **Block level data movements (Intelligent Striping)**